

Fit Time — SDSS

Parametric

Nonparametric

Foundation

n	Parametric												Nonparametric						Foundation				
	LinearGauss-Homo	LinearGauss-Hetero	Student-t	LogNormal-Homo	LogNormal-Hetero	Gamma-GLM	LinGauss-Homo-Ridge	LinGauss-Hetero-Ridge	Student-t-Ridge	LogNormal-Homo-Ridge	LogNormal-Hetero-Ridge	Gamma-GLM-Ridge	FlexCode-RF	BART-Homo	BART-Hetero	Flow-Spline	Quantile-Tree	CatMLP	MDN	TabPFN Native	TabPFN 2.5	RealTabPFN 2.5	TabICL Quantiles
500	<0.01 (<0.01)	<0.01 (<0.01)	<0.01 (<0.01)	0.02s (<0.01)	0.02s (<0.01)	1.3s (1.3s)	0.01s (<0.01)	0.01s (<0.01)	0.02s (<0.01)	0.02s (<0.01)	0.03s (<0.01)	0.29s (0.27s)	1.1s (0.01s)	4.1s (0.36s)	24.6s (2.7s)	23.7s (0.81s)	9.1s (5.8s)	2.0s (0.09s)	8.9s (4.9s)	1.1s (0.63s)	0.72s (0.33s)	0.72s (0.34s)	4.4s (2.8s)
1K	<0.01 (<0.01)	<0.01 (<0.01)	0.01s (<0.01)	0.03s (<0.01)	0.03s (<0.01)	0.05s (<0.01)	0.01s (<0.01)	0.01s (<0.01)	0.02s (<0.01)	0.04s (<0.01)	0.04s (<0.01)	0.03s (<0.01)	1.6s (0.05s)	5.3s (0.47s)	30.0s (2.8s)	39.4s (1.9s)	6.3s (2.1s)	3.1s (0.05s)	5.9s (0.29s)	0.72s (0.26s)	0.43s (<0.01)	0.42s (<0.01)	2.9s (0.06s)
10K	0.02s (<0.01)	0.02s (<0.01)	0.06s (<0.01)	0.27s (<0.01)	0.28s (<0.01)	0.23s (<0.01)	0.05s (<0.01)	0.06s (<0.01)	0.09s (<0.01)	0.30s (<0.01)	0.30s (<0.01)	0.20s (<0.01)	12.3s (0.19s)	32.2s (2.1s)	2.3m (12.5s)	4.1m (11.1s)	32.6m (32.2m)	29.3s (0.74s)	30.3s (1.0s)	3.2s (0.37s)	2.7s (0.02s)	2.8s (0.03s)	40.3s (1.6s)
50K	0.08s (<0.01)	0.09s (<0.01)	0.28s (<0.01)	1.3s (0.03s)	1.3s (0.02s)	1.0s (0.03s)	0.22s (<0.01)	0.25s (<0.01)	0.42s (0.01s)	1.4s (0.03s)	1.4s (0.02s)	0.94s (0.02s)	54.0s (0.07s)	2.9m (5.5s)	13.0m (11.8s)	11.4m (42.7s)	2.7h (47.2m)	3.2m (24.1s)	1.8m (5.0s)	36.4s (0.64s)	37.0s (0.32s)	37.2s (0.64s)	—
100K	0.15s (<0.01)	0.16s (<0.01)	0.53s (0.02s)	2.4s (0.02s)	2.5s (0.01s)	1.9s (<0.01)	0.45s (0.03s)	0.48s (<0.01)	0.79s (0.02s)	2.7s (<0.01)	2.7s (0.01s)	1.8s (0.01s)	1.9m (1.2s)	6.1m (6.4s)	26.4m (20.5s)	17.7m (5.6s)	2.1h (59.0m)	10.1m (1.5m)	3.4m (8.6s)	—	—	—	—
250K	0.38s (<0.01)	0.40s (<0.01)	1.3s (0.01s)	5.9s (0.04s)	5.9s (0.07s)	11.7s (0.38s)	1.3s (0.01s)	1.5s (<0.01)	2.3s (<0.01)	6.9s (0.05s)	7.1s (0.04s)	6.4s (0.03s)	5.7m (13.9s)	17.2m (2.4s)	—	21.7m (33.5s)	—	38.1m (7.7m)	11.1m (28.5s)	—	—	—	—
500K	0.89s (0.01s)	0.90s (<0.01)	2.9s (0.05s)	13.0s (0.44s)	13.0s (0.45s)	18.0s	3.7s	3.7s	5.4s	16.0s	16.2s	11.6s	14.7m	45.1m (45.4s)	—	28.5m (3.1m)	—	1.5h	26.7m	—	—	—	—